The documentation and process conversion measures necessary to comply with this revision shall be completed by 7 Oct 95.

INCH-POUND

MIL-S-19500/596C
7 April 1995
SUPERSEDING
MIL-S-19500/596B
17 August 1994

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT, TRANSISTOR, N-CHANNEL, SILICON, TYPES 2N7218, 2N7219, 2N7221, 2N7222, 2N7218U, 2N7219U, 2N7221U, 2N7222U, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for an N-channel, enhancement mode, MOSFET, power transistor intended for use in high density power switching applications. Four levels of product assurance are provided for each device type as specified in MIL-S-19500, with avalanche energy maximum ratings (E_{AR} and E_{AS}) and maximum avalanche current I_{AR}. Two level of product assurance for die (element evaluation).

1.2 Physical dimensions. See figure 1 (TO-254AA), figure 2 for surface mount, and figure 3 for JANHC and JANKC (die) dimensions.

1.3 Maximum ratings.

Type	V_{DSS} V_{GS} = 0 V dc, I_D = 1.0 mA dc	P_T 1/ T_C = +25°C	P_T T_A = +25°C	V_{GS}	I_{D1} 2/ T_C = +25°C	I_{D2} 2/ T_C = +100°C	I_S	I_{DM} 3/	T_{op} and T_{stg}
	V dc	W	W	V dc	A dc	A dc	A dc	A (pk)	°C
2N7218	100	125	4	±20	28	20	28	112	-55
2N7219	200	"	"	"	18	11	18	72	to
2N7221	400	"	"	"	10	6	10	40	+150
2N7222	500	"	"	"	8	5	8	32	
2N7218U	100	75		"	22	13.9	22	88	
2N7219U	200	"		"	13.9	8.8	13.9	56	
2N7221U	400	"		"	7.5	4.8	7.5	30	
2N7222U	500	"		"	6.1	3.8	6.1	24	

See footnotes at end of table.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Electronics Supply Center, ATTN: DESC-ELD, 1507 Wilmington Pike, Dayton, OH 45444-5765, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 5961

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Type	I_{AR} 2/	E_{AS}	E_{AR}	$V_{(ISO)}$ at 70,000 feet	$R_{\theta JC}$ max	$r_{DS(on)}$ max 1/ 4/ V_{GS} = 10 V dc $I_D = I_{D2}$	
						T_J = +25°C	T_J = +150°C
	A	mj	mj		°C/W	Ω	Ω
2N7218	28	250	12.5		1.0	0.077	0.154
2N7219	18	450	"		"	0.18	0.387
2N7221	10	650	"	400	"	0.55	1.32
2N7222	8	700	"	500	"	0.85	2.04
2N7218U	22	250	7.5		1.67	0.077	0.154
2N7219U	13.9	450	"		"	0.18	0.387
2N7221U	7.5	650	"	400	"	0.55	1.32
2N7222U	6.1	700	"	500	"	0.85	2.040

1/ Derate linearly 1.0 W/°C (for non-"U" suffix devices) or 0.6 W/°C (for "U" suffix devices) for $T_C > +25°C$; $P_T = \dfrac{T_J \text{ max} - T_C}{R_{\theta JC}}$

2/ $I_D = \sqrt{\dfrac{T_J \text{ max} - T_C}{(R_{\theta JC} \times R_{DS(on)} \text{ at } T_J \text{ max})}}$

3/ $I_{DM} = 4 \times I_D$ as calculated by footnote 2/.

4/ Pulsed (see 4.5.1).

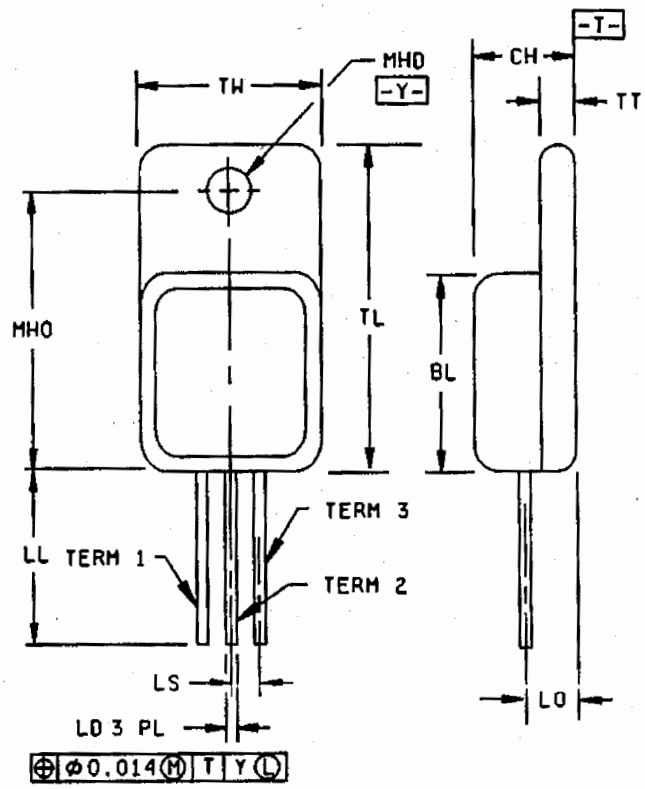
1.4 Primary electrical characteristics. Unless otherwise specified, T_C = +25°C.

Type 1/	$V_{(BR)DSS}$ minimum V_{GS} = 0 V dc I_D = 1.0 mA dc	$V_{GS(th)1}$ $V_{DS} \geq V_{GS}$ I_D = 0.25 mA		I_{DSS1} maximum V_{GS} = 0 V dc V_{DS} = 80 percent of rated V_{DS}	$r_{DS(on)1}$ V_{GS} = 10 V dc, $I_D = I_{D2}$
	V dc	V dc		μA dc	ohms
		Min	Max		
2N7218	100	2.0	4.0	25	0.077
2N7219	200	2.0	4.0	25	0.18
2N7221	400	2.0	4.0	25	0.55
2N7222	500	2.0	4.0	25	0.85

1/ Electrical characteristics for "U" suffix devices are identical to the corresponding non-"U" suffix devices.

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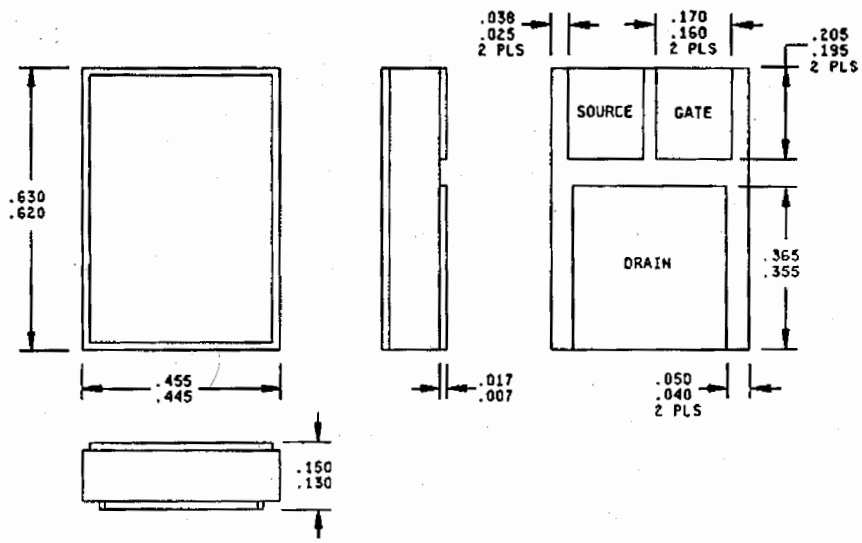
Ltr	Dimensions			
	Inches		Millimeters	
	Min	Max	Min	Max
CH	.249	.260	6.32	6.60
TT	.040	.050	1.02	1.27
LO	.150 BSC		3.81	
TL	.790	.800	20.07	20.32
BL	.535	.545	13.59	13.84
LS	.150 BSC		3.81	
TW	.535	.545	13.59	13.84
LL	1.195	1.235	30.35	31.37
MHO	.665	.685	16.89	17.40
MHD	.139	.149	3.53	3.78
LD	.035	.045	0.89	1.14
Term 1	Drain			
Term 2	Source			
Term 3	Gate			



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Glass meniscus included in dimensions D and E.
4. All terminals are isolated from case.
5. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.

FIGURE 1. Dimensions and configuration (TO-254AA).

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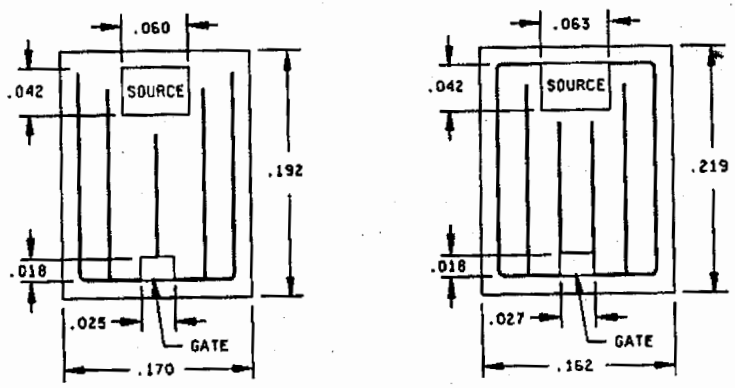


Inches	mm	Inches	mm
.007	.18	.170	4.32
.017	.43	.195	4.95
.025	.64	.205	5.21
.038	.97	.355	9.02
.040	1.02	.365	9.27
.050	1.27	.445	11.30
.130	3.30	.455	11.56
.150	3.81	.620	15.75
.160	4.06	.630	16.00

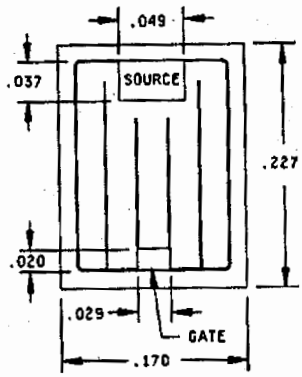
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Dimensions and configuration of surface mount package outline.

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2N7218 2N7219



2N7221 AND 2N7222

Inches	mm	Inches	mm	Inches	mm
.018	0.46	.037	0.94	.162	4.11
.020	0.51	.042	1.07	.170	4.32
.025	0.64	.049	1.24	.192	4.88
.027	0.69	.060	1.52	.219	5.56
.029	0.74	.063	1.60	.227	5.77

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 inch (0.13 mm).
4. The physical characteristics of the die thickness are .0187 inch (0.475 mm). The back metals are chromium, nickel, and silver. The top metal is aluminum and the back contact is the drain.

FIGURE 3. JANHC and JANKC (A-version) die dimensions.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Defense Printing Service Detachment Office, Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated detail specification. The individual item requirements shall be in accordance with MIL-S-19500 and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500.

E_{AS}	Single pulse avalanche energy capability.
E_{AR}	Repetitive avalanche energy capability.
I_{AR}	Rated avalanche current, repetitive and non-repetitive.
$V_{(ISO)}$	Source pin to case isolation voltage.
$I_{(ISO)}$	Source pin to case isolation current.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and on figure 1 herein. Methods used for electrical isolation of the terminal feedthroughs shall employ materials that contain a minimum of 90 percent AL_2O_3 (ceramic). Examples of such construction techniques are metallized ceramic eyelets or ceramic walled packages.

3.3.1 Lead formation, material, and finish. Lead material shall be Kovar or Alloy 52; a copper core or plated core is permitted. Lead finish shall be solderable in accordance with MIL-STD-750 and MIL-S-19500. Where a choice of finish is desired, it shall be specified in the acquisition document (see 6.2). When lead formation is performed, as a minimum, the vendor shall perform 100 percent hermetic seal in accordance with screen 14 and 100 percent dc testing in accordance with group A, subgroup 2 herein.

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3.3.2 Internal construction. Multiple chip construction shall not be permitted.

3.4 Marking. Marking shall be in accordance with MIL-S-19500, except at the option of the manufacturer, the country of origin or the manufacturers identification may be omitted from the body of the transistor.

3.5 Electrostatic discharge protection. The devices covered by this specification require electrostatic protection.

3.5.1 Handling. MOS devices must be handled with certain precautions to avoid damage due to the accumulation of static charge. The following handling procedures shall be followed:

a. Devices shall be handled on benches with conductive handling devices.

b. Ground test equipment, tools, and personnel handling devices.

c. Do not handle devices by the leads.

d. Store devices in conductive foam or carriers.

e. Avoid use of plastic, rubber, or silk in MOS areas.

f. Maintain relative humidity above 50 percent, if practical.

g. Care shall be exercised, during test and troubleshooting, to apply not more than maximum rated voltage to any lead.

h. Gate must be terminated to source. $R \leq 100$ k, whenever bias voltage is to be applied drain to source.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein. Alternate flow is allowed for qualification inspection in accordance with figure 2 of MIL-S-19500.

4.2.1 Group E inspection. Group E inspection shall be in accordance with table II herein.

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4.3 Screening (JANS, JANTX, and JANTXV levels). Screening shall be in accordance with table II of MIL-S-19500, and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table II of MIL-S-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
1/	Gate stress test (see 4.5.5)	Gate stress test (see 4.5.5)
1/	Method 3470 (see 4.5.4)	Method 3470 (see 4.5.4)
1/	Method 3161 (see 4.5.3)	Method 3161 (see 4.5.3)
9 1/	I_{GSS1}, I_{DSS1}, subgroup 2 of table I herein	Subgroup 2 of table I herein
10	Method 1042, test condition B	Method 1042, test condition B
11	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$ subgroup 2 of table 1 herein; ΔI_{GSS1} = ±20 nA dc or 100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater.	I_{GSS1}, I_{DSS1}, $r_{DS(on)1}$, $V_{GS(th)1}$, subgroup 2 of table I herein.
12	Method 1042, test condition A	Method 1042, test condition A or T_A = +175°C and t = 48 hours.
13	Subgroups 2 and 3 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.	Subgroup 2 of table I herein; ΔI_{GSS1} = ±20 nA dc or ±100 percent of initial value, whichever is greater. ΔI_{DSS1} = ±25 μA dc or ±100 percent of initial value, whichever is greater. $\Delta r_{DS(on)(1)}$ = ±20 percent of initial value. $\Delta V_{GS(th)1}$ = ±20 percent of initial value.

1/ Shall be performed anytime before screen 10.

4.3.1 Screening (JANHC and JANKC). Screening of die shall be in accordance with MIL-S-19500, appendix H. As a minimum die, shall be 100 percent probed in accordance with group A, subgroup 2 except test current shall not exceed 20 amperes.

4.3.2 JANHC and JANKC die. Qualification shall be in accordance with MIL-S-19500.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500 and table I herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500 and herein. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

Subgroup	Method	Condition
3	1051	Test condition G.
3	2037	Test condition A. All internal bond wires for each device shall be pulled separately.
4	1042	Test condition D; the heating cycle shall be 1 minute minimum, 2,000 cycles. No heat sink nor forced air cooling on the device shall be permitted during the cycle.
5	1042	Accelerated steady-state operation life; test condition A, V_{DS} = rated T_A = +175°C, t = 120 hours minimum. Read and record $V_{(BR)DSS}$ (pre and post) at 1 mA = I_D. Read and record I_{DSS} (pre and post). Deltas for $V_{(BR)DSS}$ shall not exceed 10 percent and I_{DSS} shall not exceed 25 μA. Accelerated steady-state gate stress; condition B, V_{GS} = rated, T_A = +175°C, t = 24 hours.
5	2037	Bond strength (Al-Au die interconnects only); test condition A.
6	3161	See 4.5.2.

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

Subgroup	Method	Condition
2	1051	Test condition G.
3	1042	Test condition D, 2,000 cycles minimum. The heating cycle shall be 1 minute minimum.
4	2075	See 3.3.2.
5, 6, and 7		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500 and as follows. Electrical measurements (end-points) shall be in accordance with the inspections of table I, group A, subgroup 2 herein.

Subgroup	Method	Condition
2	1056	Test condition B.
2	2036	Tension: Test condition A; weight = 10 pounds, t = 10 s (not applicable to "U" suffix version).
6	1042	Test condition D, 6,000 cycles minimum. The heating cycle shall be 1 minute minimum.

4.5 Methods of inspection. Methods of inspection shall be as specified in appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurements shall be as specified in MIL-STD-750.

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4.5.2 Thermal resistance. Thermal resistance measurements shall be performed in accordance with method 3161 of MIL-STD-750. The maximum limit of $R_{\Theta JC(max)}$ shall be 1.0°C/W for devices in the TO-254AA configuration. The maximum limit of $R_{\Theta JC(max)}$ for surface mount devices shall be 1.67°C/W. The following parameter measurements shall apply:

a. Measuring current (I_M): 10 mA.

b. Drain heating current (I_H): 3.3 A (2 A minimum for surface mount).

c. Heating time (t_H): Steady state (see MIL-STD-750, method 3161 for definition).

d. Drain-source heating voltage (V_H): 25 V minimum.

e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.

f. Sample window time (t_{SW}): 10 μs maximum.

4.5.3 Thermal impedance ($Z_{\Theta JC}$ measurements). The $Z_{\Theta JC}$ measurements shall be performed in accordance with MIL-STD-750, method 3161. The maximum limit (not to exceed figure 4, thermal impedance curves and the group A, subgroup 2 limits) for $Z_{\Theta JC}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition. This procedure may be used in lieu of an inline process monitor.

a. Measuring current (I_M): 10 mA.

b. Drain heating current (I_H): 3.3 A (2 A minimum for surface mount).

c. Heating time (t_H): 100 ms (50 ms for surface mount).

d. Drain-source heating voltage (V_H): 25 V dc.

e. Measurement time delay (t_{MD}): 30 to 60 μs maximum.

f. Sample window time (t_{SW}): 10 μs maximum.

4.5.4 Single pulse avalanche energy (E_{AS}).

a. Peak current (I_{AS}): I_{D1}.

b. Peak gate voltage (V_{GS}): 10 V dc.

c. Gate to source resistor (R_{GS}): $25 \leq R_{GS} \leq 200\ \Omega$.

d. Initial case temperature: +25°C +10°C, -5°C.

e. Inductance: $(2\ E_{AS}/(I_{D1})^2)\ ((V_{BR} - V_{DD})/V_{BR})$ mH minimum.

f. Number of pulses to be applied: 1 pulse minimum.

g. Supply voltage (V_{DD}): 50 V dc, 25 V dc for 100 V dc devices.

4.5.5 Gate stress test.

V_{GS} ± 30 V dc minimum.

t = 250 μs minimum.

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TABLE I. Group A inspection.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical inspection	2071					
Subgroup 2						
Thermal impedance 3/	3161	See 4.5.3	$Z_{\Theta JC}$		0.9	°C/W
Breakdown voltage, drain to source	3407	V_{GS} = 0 V dc, I_D = 1.0 mA, bias condition C	$V_{(BR)DSS}$			
2N7218				100		V dc
2N7219				200		V dc
2N7221				400		V dc
2N7222				500		V dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)1}$	2.0	4.0	V dc
Gate current	3411	V_{GS} = +20 V dc, V_{DS} = 0 V dc, bias condition C	I_{GSSF1}		100	nA dc
Gate current	3411	V_{GS} = -20 V dc, V_{DS} = 0 V dc, bias condition C	I_{GSSR1}		-100	nA dc
Drain current	3413	V_{GS} = 0 V dc, bias condition C, V_{DS} = 80 percent of rated V_{DS}	I_{DSS1}		25	µA dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A pulsed, (see 4.5.1), I_D = rated I_{D2} (see 1.3)	$r_{DS(on)1}$			
2N7218					0.077	Ω
2N7219					0.18	Ω
2N7221					0.55	Ω
2N7222					0.85	Ω
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc, condition A pulsed, (see 4.5.1), I_D = rated I_{D1} (see 1.3)	$r_{DS(on)2}$			
2N7218					0.125	Ω
2N7219					0.25	Ω
2N7221					0.70	Ω
2N7222					0.95	Ω

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 2 - Continued						
Forward voltage (source drain diode)	4011	V_{GS} = 0 V dc, I_D = rated I_{D1} (see 1.3)	V_{SD}			
2N7218					1.5	V
2N7219					1.5	V
2N7221					1.5	V
2N7222					1.5	V
Subgroup 3						
High temperature operation:		$T_C = T_J$ = +125°C				
Gate current	3411	Bias condition C, V_{GS} = +20 V dc, V_{DS} = 0 V dc	I_{GSSF2}		200	nA dc
Gate current	3411	Bias condition C, V_{GS} = -20 V dc, V_{DS} = 0 V dc	I_{GSSR2}		-200	nA dc
Drain current	3413	Bias condition C, V_{GS} = 0 V dc, V_{DS} = 100 percent of rated V_{DS}	I_{DSS2}		1.0	mA dc
		V_{DS} = 80 percent of rated V_{DS}	I_{DSS3}		0.25	mA dc
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)2}$	1.0		V dc
Static drain to source on-state resistance	3421	V_{GS} = 10 V dc pulsed, (see 4.5.1), $I_D = I_{D2}$ (see 1.3)	$r_{DS(on)3}$			
2N7218					0.24	Ω
2N7219					0.48	Ω
2N7221					1.44	Ω
2N7222					2.04	Ω
Low temperature operation:		$T_C = T_J$ = -55°C				
Gate to source voltage (threshold)	3403	$V_{DS} \geq V_{GS}$, I_D = 0.25 mA	$V_{GS(th)3}$		5.0	V dc

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Switching time test	3472	I_D = rated I_{D2}, V_{GS} = 10 V dc pulsed, Gate drive impedance = 9.1 Ω, V_{DD} = 50 percent of $V_{BR(DSS)}$				
Turn-on delay time			$t_{d(on)}$			
2N7218					21	ns
2N7219					20	ns
2N7221					25	ns
2N7222					21	ns
Rise time			t_r			
2N7218					105	ns
2N7219					105	ns
2N7221					92	ns
2N7222					73	ns
Turn-off delay time			$t_{d(off)}$			
2N7218					64	ns
2N7219					58	ns
2N7221					79	ns
2N7222					72	ns
Fall time			t_f			
2N7218					65	ns
2N7219					67	ns
2N7221					58	ns
2N7222					51	ns
Subgroup 5						
Safe operating area test (high voltage)	3474	See figure 5 (derate by a factor of 1.67 for "U" suffix devices), t_p = 10 ms, V_{DS} = 80 percent of rated $V_{BR(DSS)}$, V_{DS} = 200 V dc maximum				
Electrical measurements		See table I, group A, subgroup 2				
Subgroup 6						
Not applicable						

See footnotes at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/ 2/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Gate charge	3471	Condition B				
On-state gate charge			$Q_{g(on)}$			
2N7218					59	nC
2N7219					60	nC
2N7221					65	nC
2N7222					68.5	nC
Gate to source charge	3053		Q_{gs}			
2N7218					12	nC
2N7219					10.6	nC
2N7221					10	nC
2N7222					12.5	nC
Gate to drain charge			Q_{gd}			
2N7236					30.7	nC
2N7219					37.6	nC
2N7221					40.5	nC
2N7222					42.4	nC
Reverse recovery time	3473	di/dt = 100 A/μs, $V_{DD} \leq 30$ V dc, I_D = rated I_{D1} (see 1.3)	t_{rr}			
2N7218					400	ns
2N7219					500	ns
2N7221					600	ns
2N7222					700	ns

1/ For sampling plan, see MIL-S-19500.

2/ Unless otherwise specified, inspections, conditions, and limits for "U" suffix devices are identical to non-"U" suffix devices.

3/ This test is required for the following end-point measurements only (not intended for screen 13):

JANS - group B, subgroups 3 and 4

JAN, JANTX and JANTXV - group B, subgroups 2 and 3;
group C, subgroup 6;
group E, subgroup 1

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TABLE II. Group E inspection (all quality levels) for qualification only.

Inspection	MIL-STD-750		Qualification inspection
	Method	Conditions	
Subgroup 1			45 devices, c = 1
Temperature cycling	1051	-55°C to +150°C, 500 cycles	
Hermetic seal	1071		
Fine leak Gross leak			
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 2 1/			45 devices, c = 1
Steady-state reverse bias	1042	Condition A; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Steady-state gate bias	1042	Condition B; 1,000 hours	
Electrical measurements		See table I, group A, subgroup 2	
Subgroup 3			3 devices, c = 0
Destructive physical analysis	2101		
Subgroup 4			5 devices, c = 0
Thermal resistance	3161	See 4.5.2	
Subgroup 5 2/			
Barometric Pressure test	1001	Condition C, $V_{(ISO)} = V_{DS}$	5 devices, c = 0
2N7221 2N7222		V_{DS} = 400 V dc V_{DS} = 500 V dc	
Subgroup 6			5 devices, c = 0
Repetitive avalanche energy	3469	a. Peak current (I_{AR}): I_D. b. Peak gate voltage (V_{GS}): -10 V dc. c. Gate to source resistors: $25 \leq R_{GS} \leq 200$. d. Temperature: T_J = +150°C, +0°C, -10°C. e. Inductance: $(2^{E_{AR}}/(I_{D1})^2)$ x $((V_{BR} - V_{DD})/V_{BR})$ mH minimum. f. Number of pulses to be applied: 3.6×10^8. g. Supply voltage (V_{DD}): 50 V dc. h. Time in avalanche: 2 µs minimum, 20 µs maximum i. Frequency: 500 Hz minimum.	
Electrical measurements		See table I, group A, subgroup 2	

1/ A separate sample for each test may be pulled.
2/ Not required for 2N7218, 2N7218U, 2N7219, and 2N7219U).
3/ Unless otherwise specified, inspections, conditions, and limits for "U" suffix devices are identical to non-"U" suffix devices.

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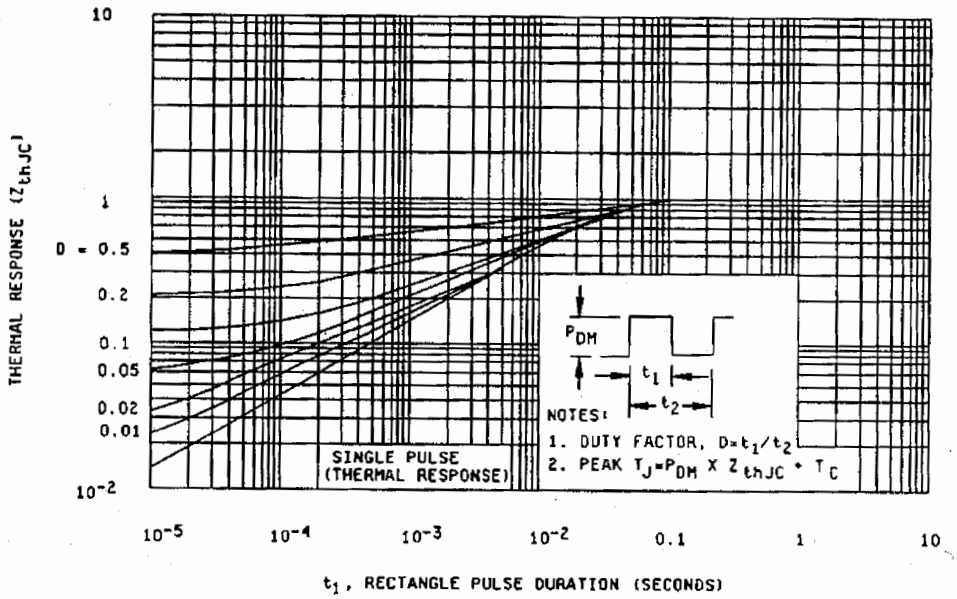


FIGURE 4. Thermal response.

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2N7218

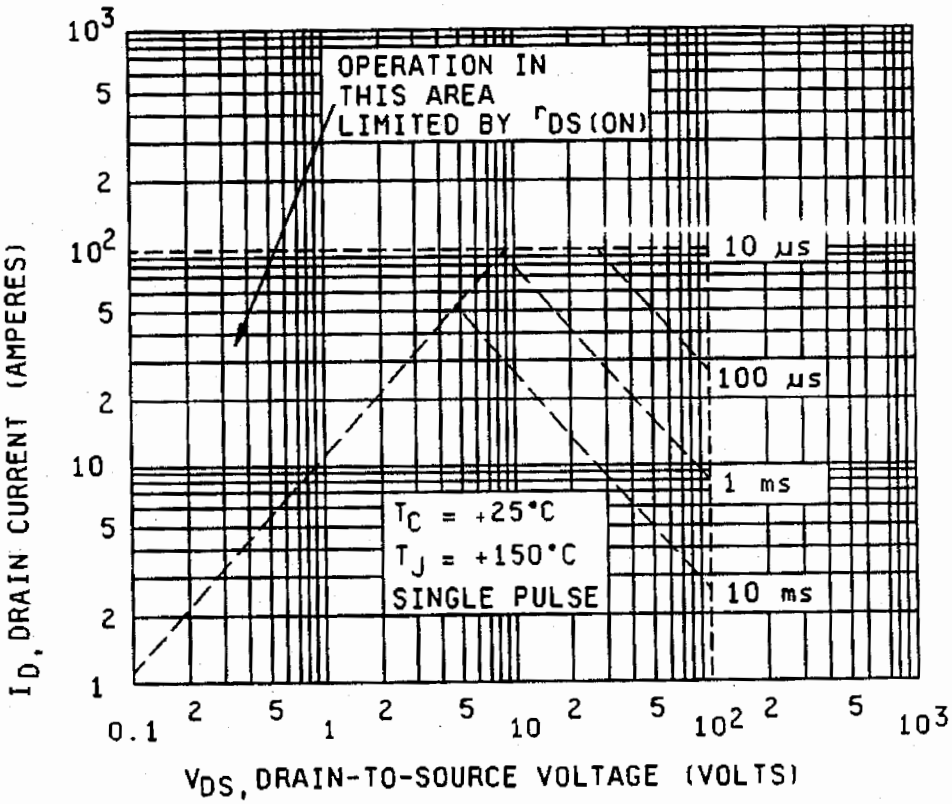


FIGURE 5. Safe operating area.

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2N7219

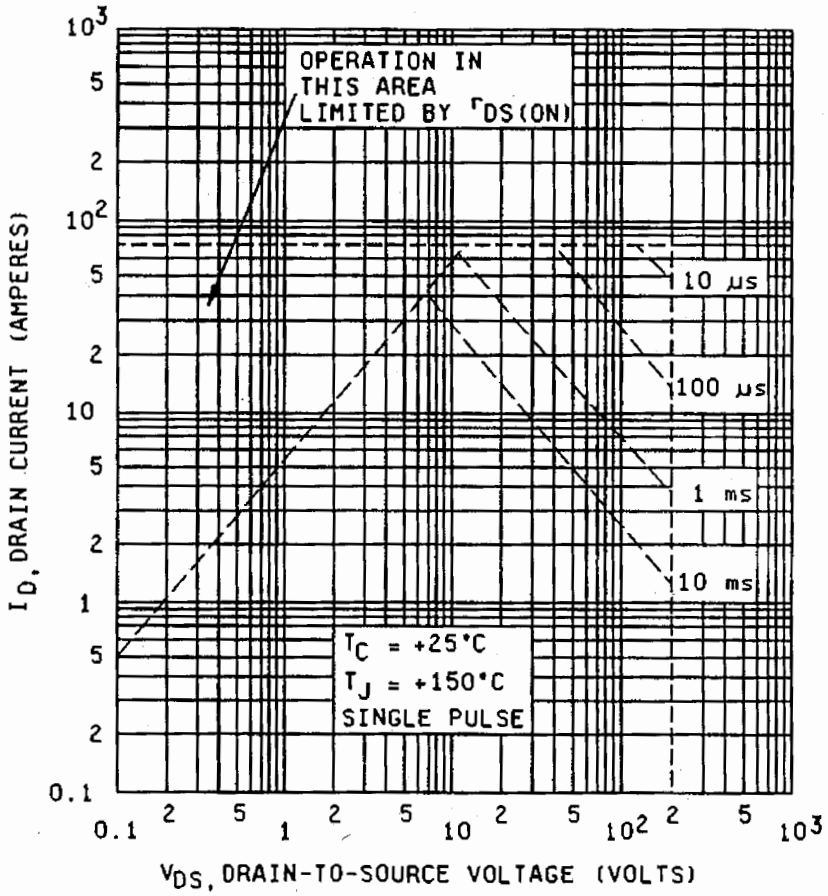


FIGURE 5. Safe operating area - Continued.

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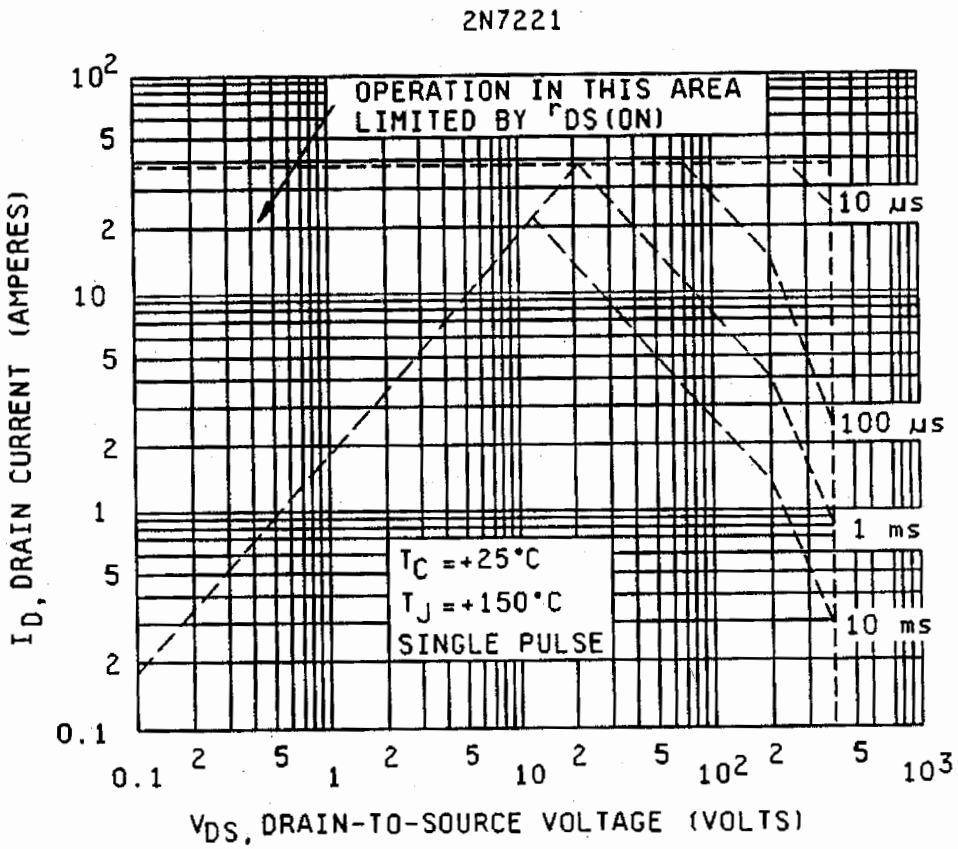


FIGURE 5. Safe operating area - Continued.

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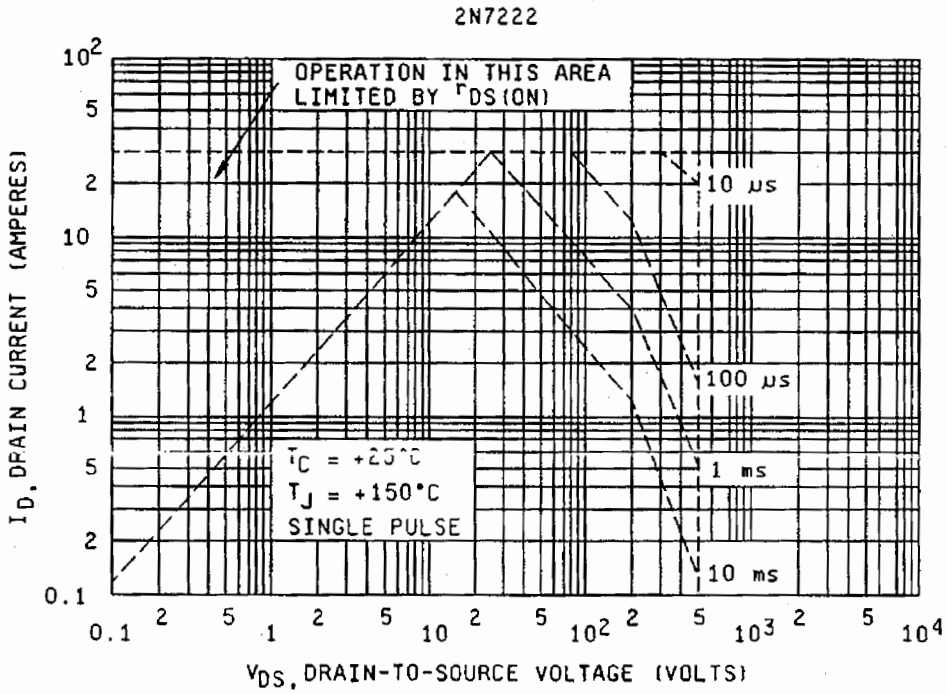


FIGURE 5. Safe operating area - Continued.

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead formation and finish as specified (see 3.3.1).

d. Product assurance level and type designator.

e. For die acquisition, specify the JANHC or JANKC letter version (see figures 3, 4, and 5).

6.3 Substitution information. Devices covered by this specification are substitutable for the manufacturer's and user's PIN. This information in no way implies that manufacturer's PIN's are suitable as a substitute for the military PIN.

Military PIN	Manufacturer's CAGE	Manufacturer's and user's PIN
2N7218	59993	IRFM 140
2N7219	59993	IRFM 240
2N7221	59993	IRFM 340
2N7222	59993	IRFM 440
2N7218U	59993	IRFN 140
2N7219U	59993	IRFN 240
2N7221U	59993	IRFN 340
2N7222U	59993	IRFN 440

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6.4 Suppliers of JANHC and JANKC die. The qualified die suppliers with the applicable letter version (example, JANHCA2N7218) will be identified on the QPL.

JANC ordering information		
	Manufacturer	
PIN	59993	
2N7218	JANHCA2N7218 JANKCA2N7218	
2N7219	JANHCA2N7219 JANKCA2N7219	
2N7221	JANHCA2N7221 JANKCA2N7221	
2N7222	JANHCA2N7222 JANKCA2N7222	

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - ES

(Project 5961-1765)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete blocks 4, 5, 6, and 7.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-19500/596C	2. DOCUMENT DATE (YYMMDD) 95/04/07

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, REPETITIVE AVALANCHE, FIELD EFFECT, TRANSISTOR, N-CHANNEL, SILICON, TYPES 2N7218, 2N7219, 2N7221, 2N7222, 2N7218U, 2N7219U, 2N7221U, 2N7222U, JAN, JANTX, JANTXV, JANS, JANHC, AND JANKC

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial 513-296-6048	(2) AUTOVON 986-6048
c. ADDRESS (Include Zip Code) Defense Electronics Supply Center Attn: DESC-ELDT Dayton, Ohio 45444-5765	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

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